



US 20250279932A1

(19) **United States**

(12) **Patent Application Publication**
Singh et al.

(10) **Pub. No.: US 2025/0279932 A1**

(43) **Pub. Date: Sep. 4, 2025**

(54) **GENERATING AND UTILIZING
PROVISIONING TEMPLATES TO
PROVISION VOICE, VIDEO, AND DATA
COMMUNICATION SERVICES**

(71) Applicant: **AT&T Intellectual Property I, L.P.**,
Atlanta, GA (US)

(72) Inventors: **Nishan Singh**, Freehold, NJ (US);
Abdulla Udaipurwala, Monroe
Township, NJ (US); **Stanimir Bonev**,
Manalapan, NJ (US); **Subhash Kapoor**,
Wall, NJ (US)

(73) Assignee: **AT&T Intellectual Property I, L.P.**,
Atlanta, GA (US)

(21) Appl. No.: **19/206,288**

(22) Filed: **May 13, 2025**

Related U.S. Application Data

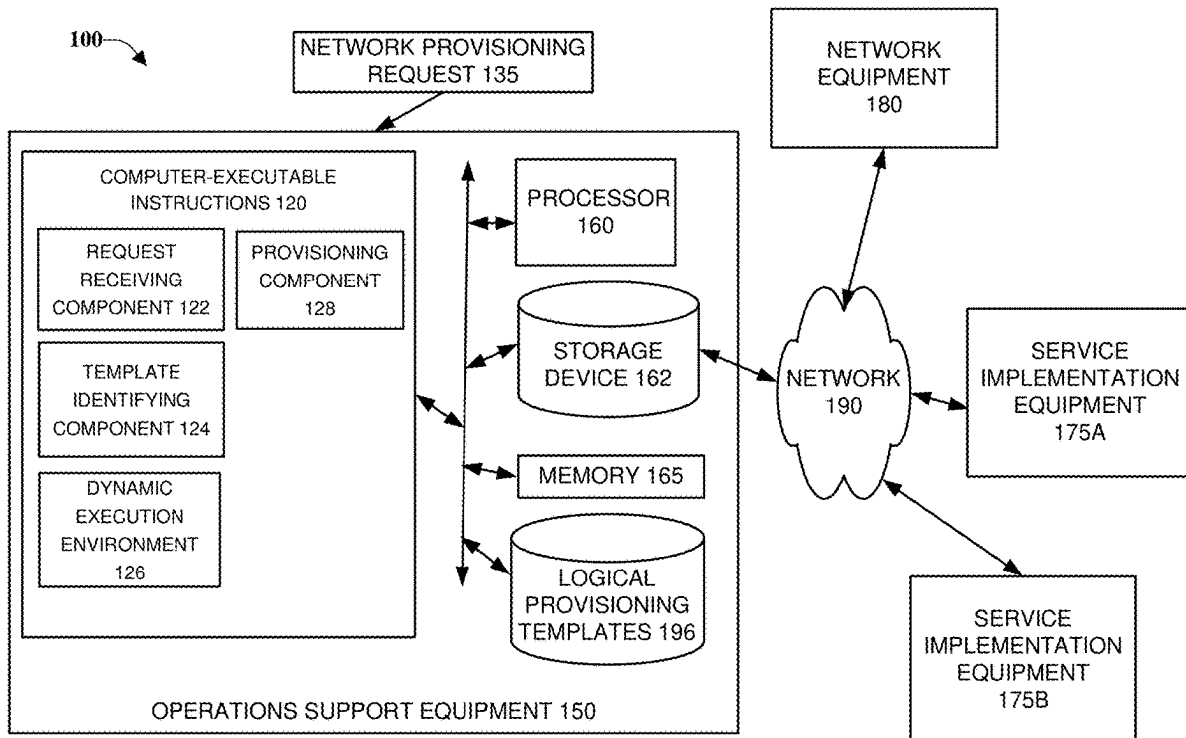
(63) Continuation of application No. 18/307,771, filed on
Apr. 26, 2023, now Pat. No. 12,328,225, which is a
continuation of application No. 17/355,396, filed on
Jun. 23, 2021, now Pat. No. 11,671,316.

Publication Classification

(51) **Int. Cl.**
H04L 41/0806 (2022.01)
G06F 11/30 (2006.01)
G06F 11/34 (2006.01)
H04L 12/66 (2006.01)
H04L 41/5041 (2022.01)
H04L 45/50 (2022.01)
(52) **U.S. Cl.**
CPC H04L 41/0806 (2013.01); **G06F 11/3006**
(2013.01); **G06F 11/3409** (2013.01); **H04L**
12/66 (2013.01); **H04L 41/5048** (2013.01);
H04L 45/50 (2013.01)

(57) **ABSTRACT**

The technologies described herein are generally directed to provisioning services from a network. For example, a method described herein can include facilitating receiving a provisioning request for a network provisioning task. Further, the method can include, based on the network provisioning task, selecting a logical provisioning template to facilitate a performance of the network provisioning task. Further, based on the logical provisioning template, the method can include identifying a programming object comprising logical commands for the performance of the network provisioning task. The method can include executing the logical commands of the programming object to control service implementation equipment to perform the network provisioning task.



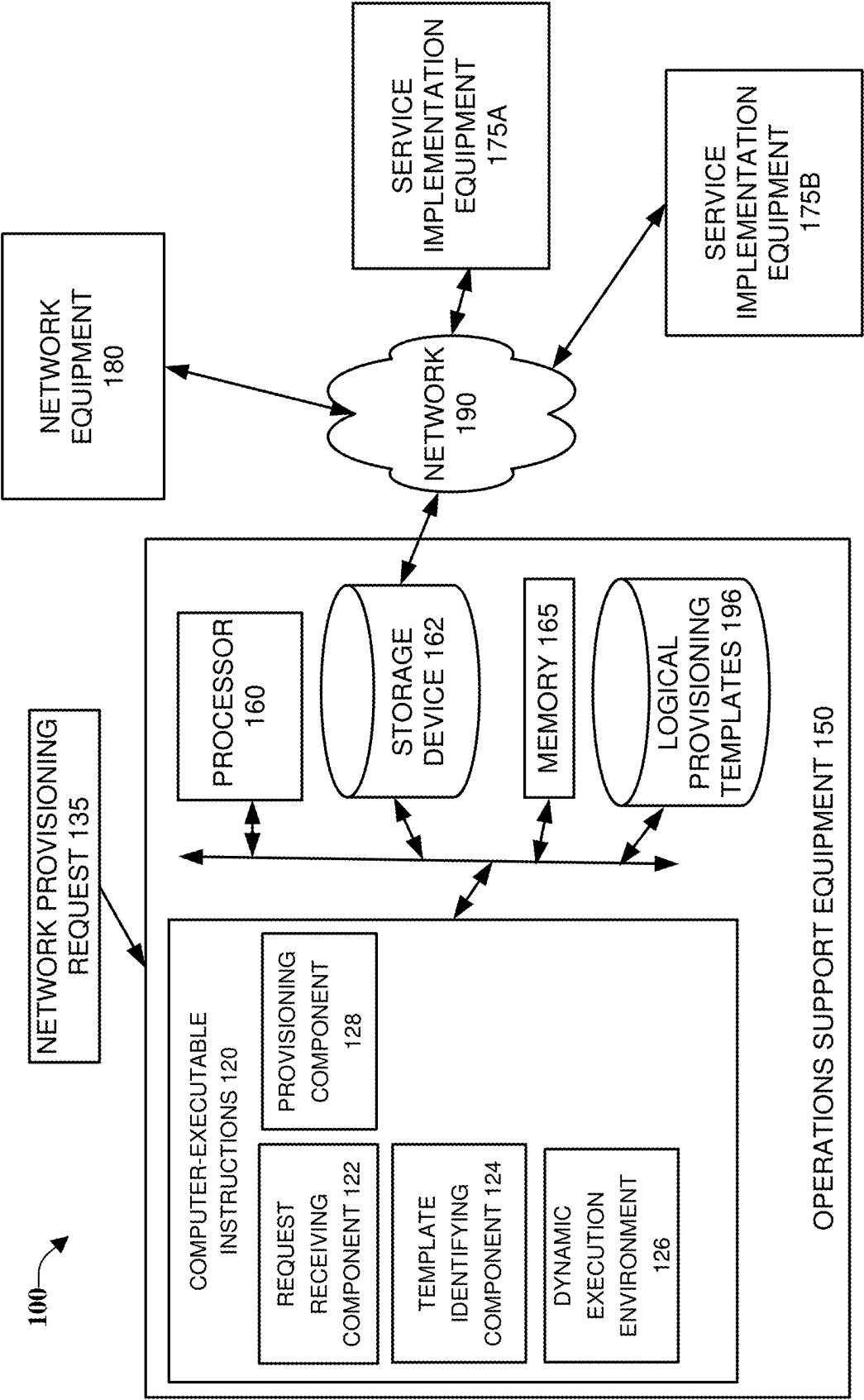


FIG. 1

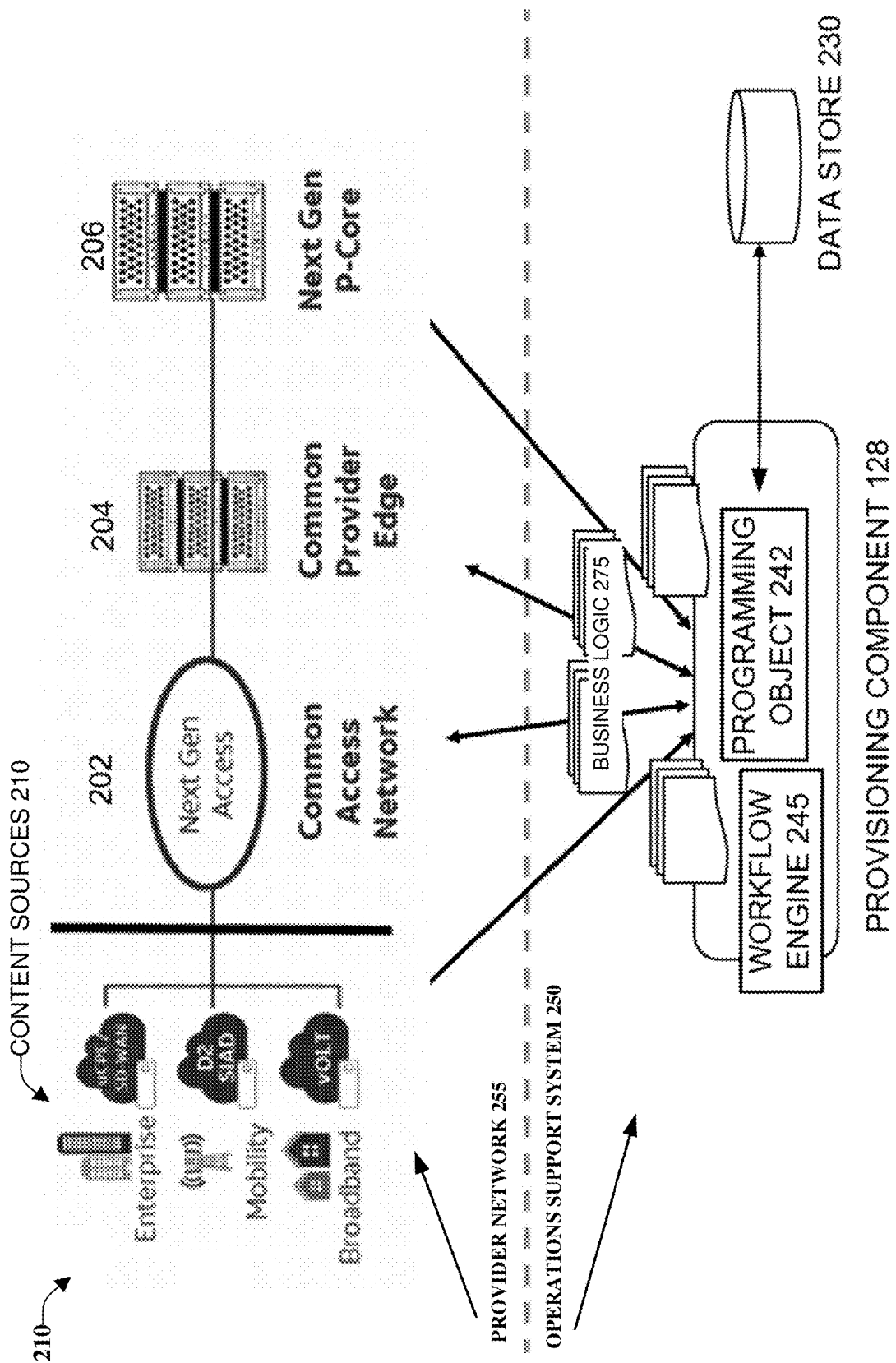


FIG. 2

300

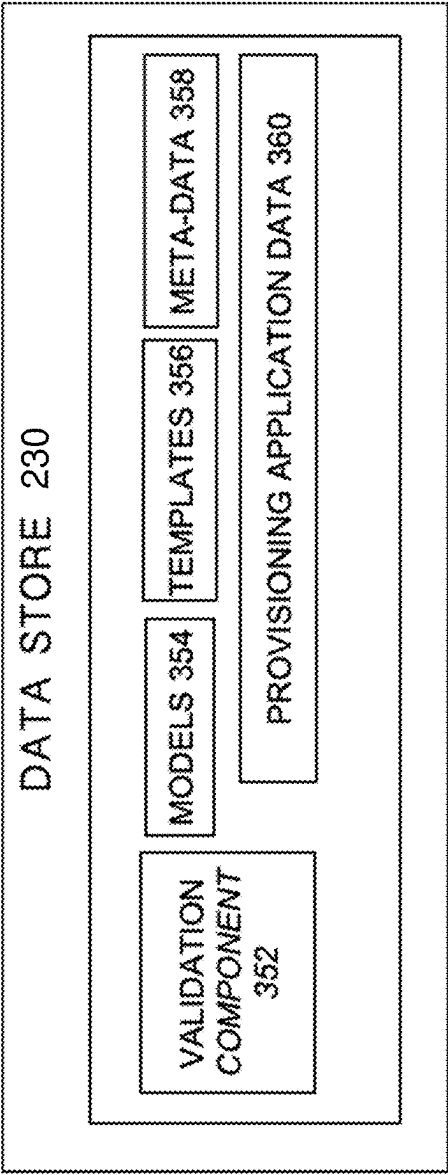


FIG. 3

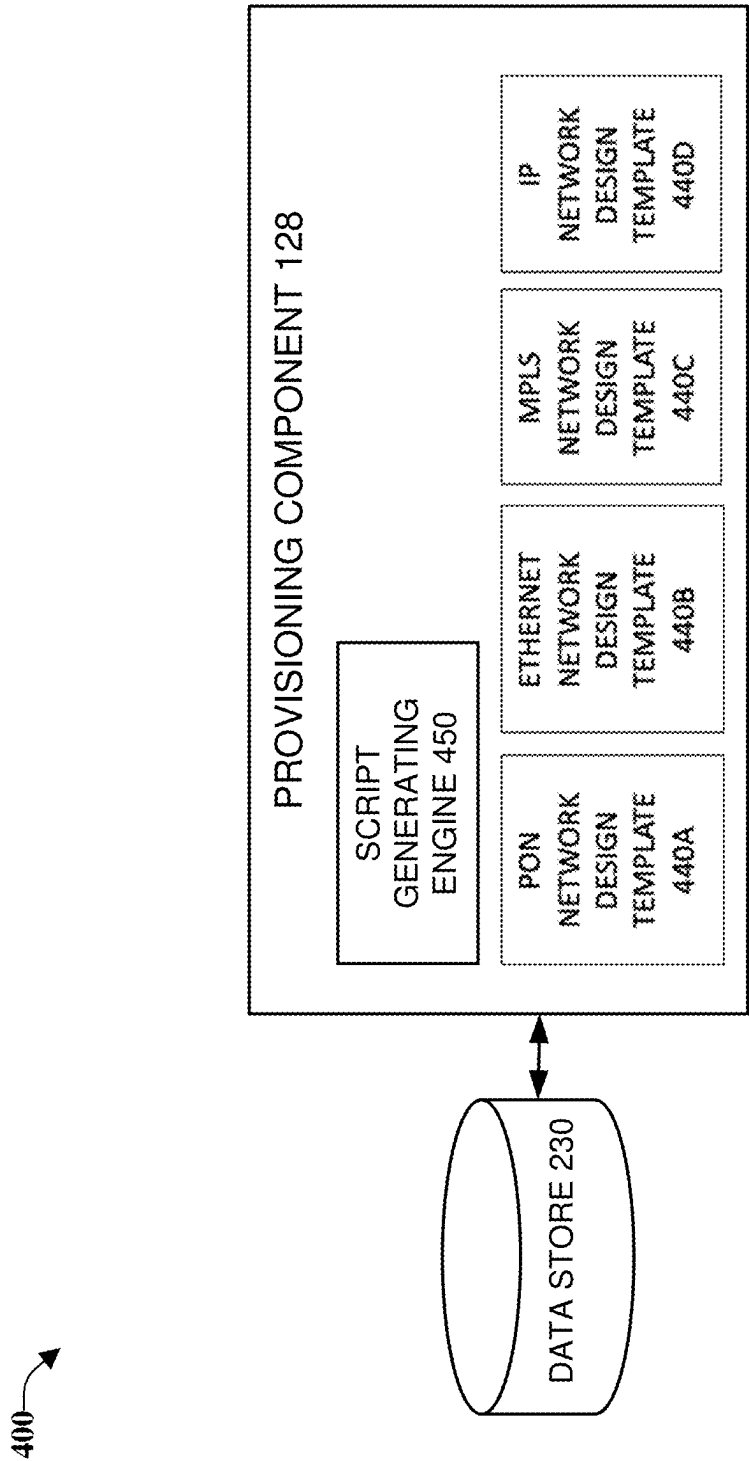


FIG. 4

500

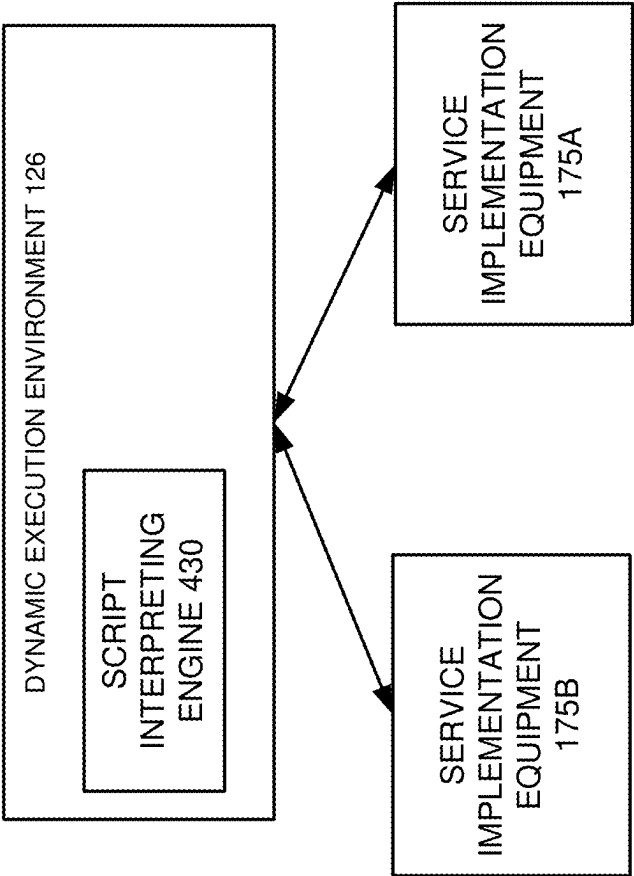
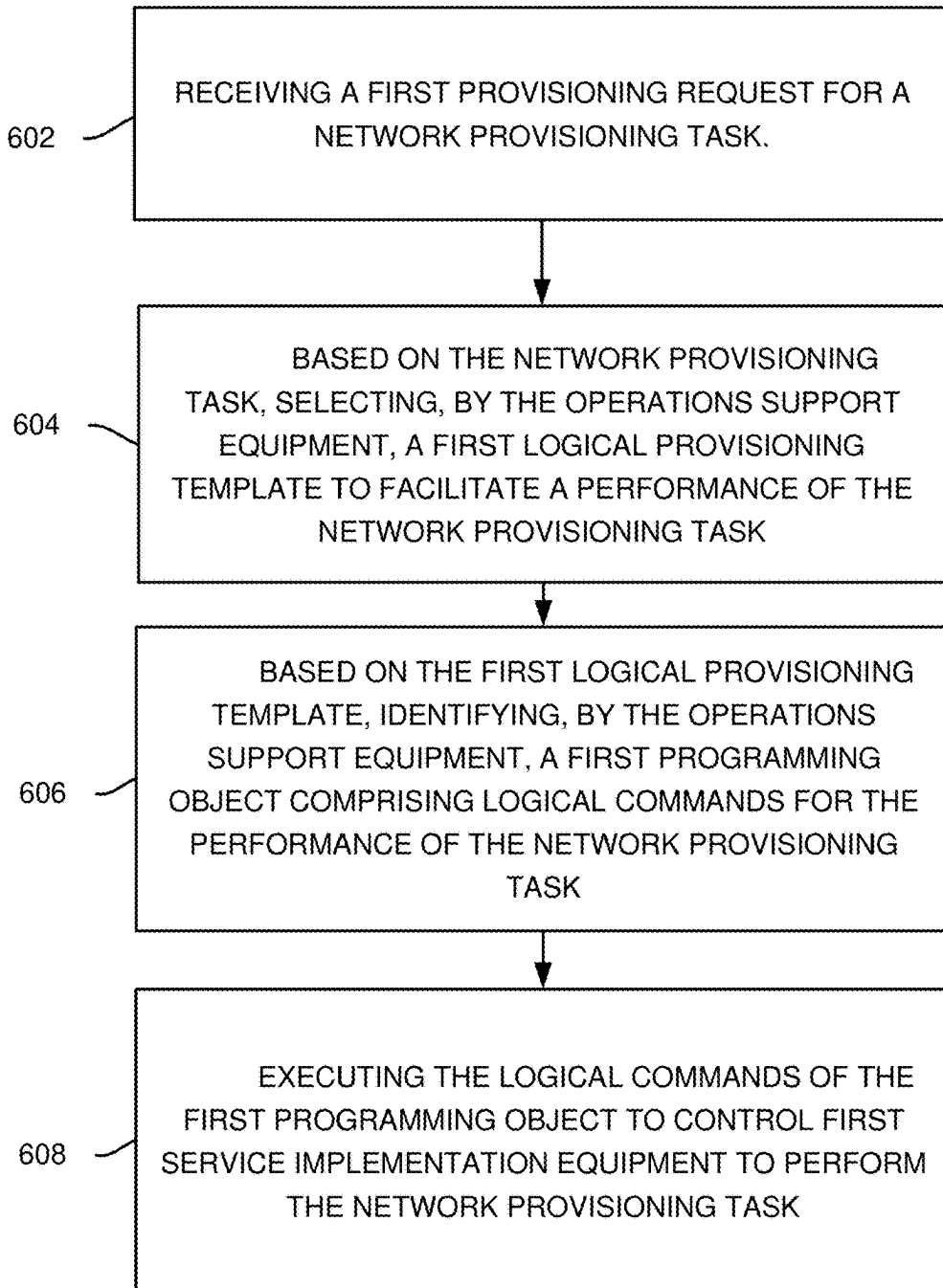
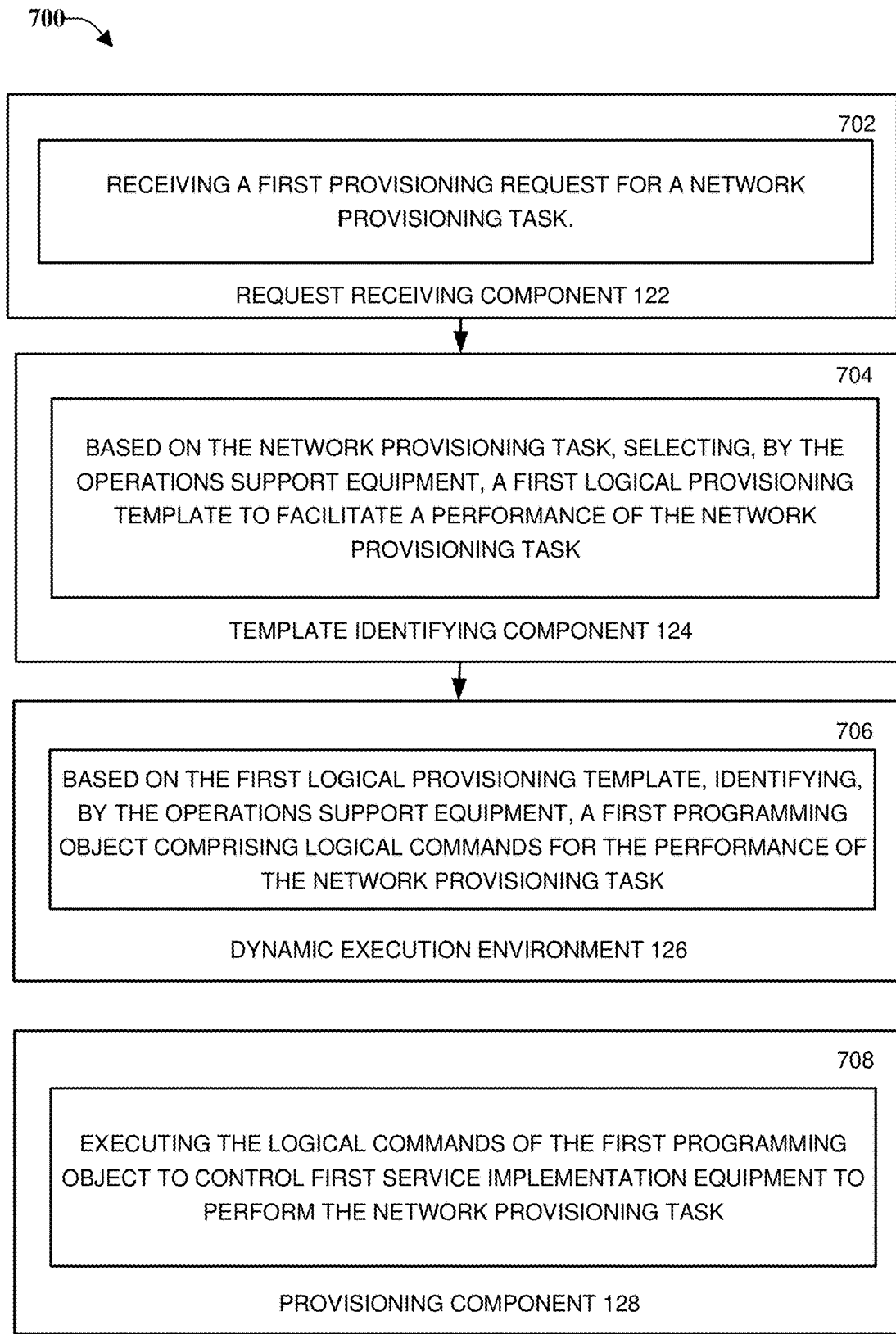
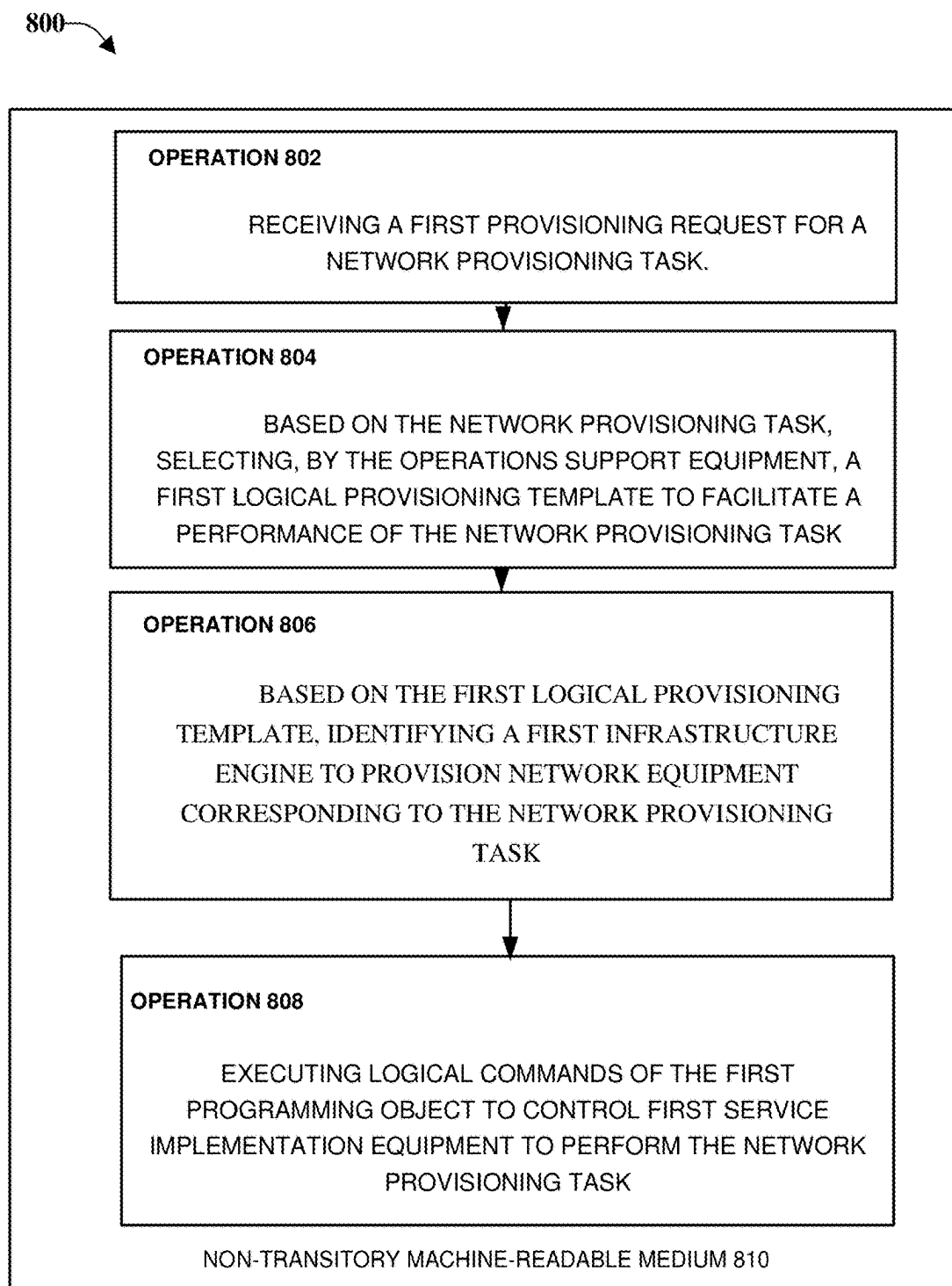


FIG. 5

600

**FIG. 6**

**FIG. 7**

**FIG. 8**

900

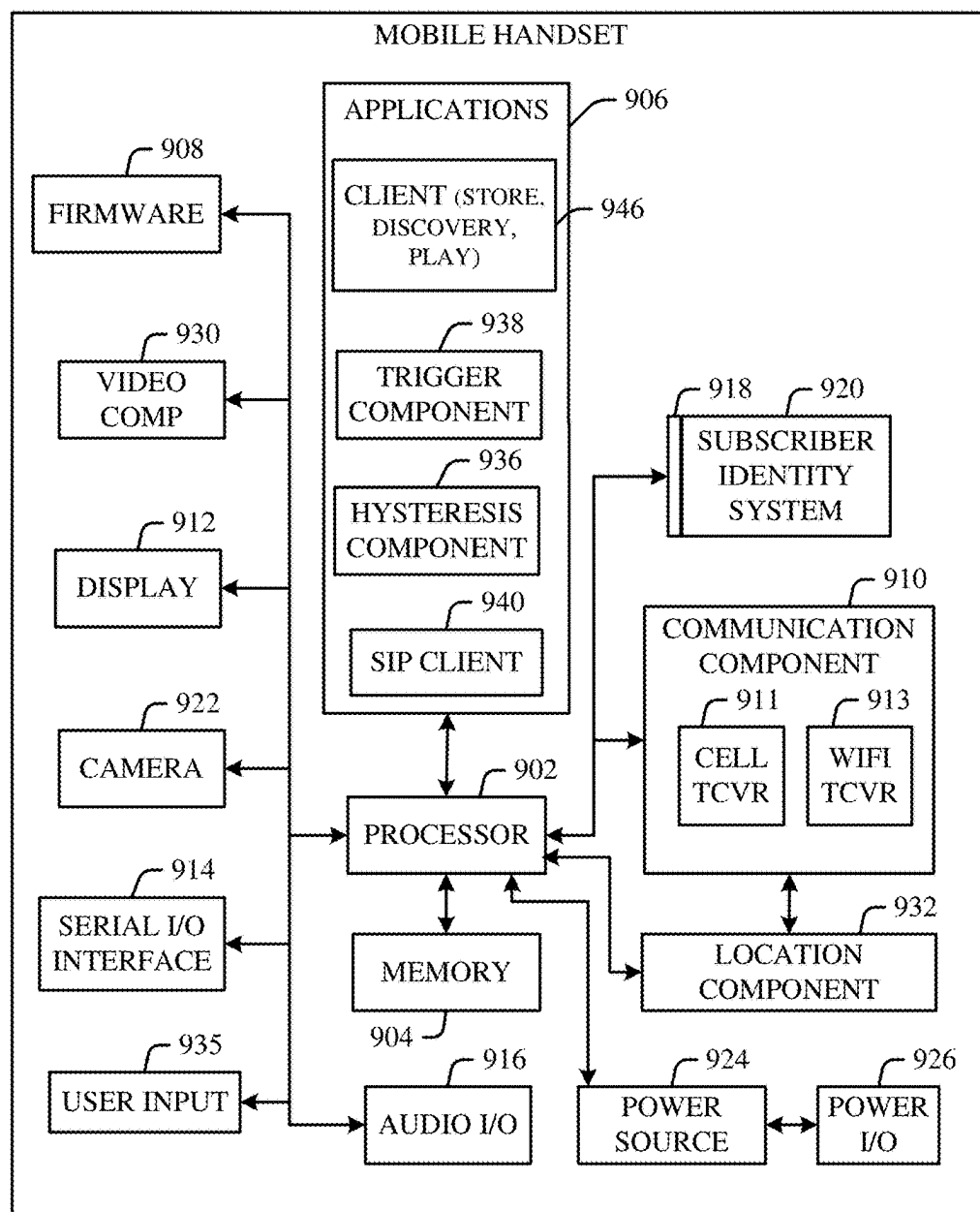


FIG. 9

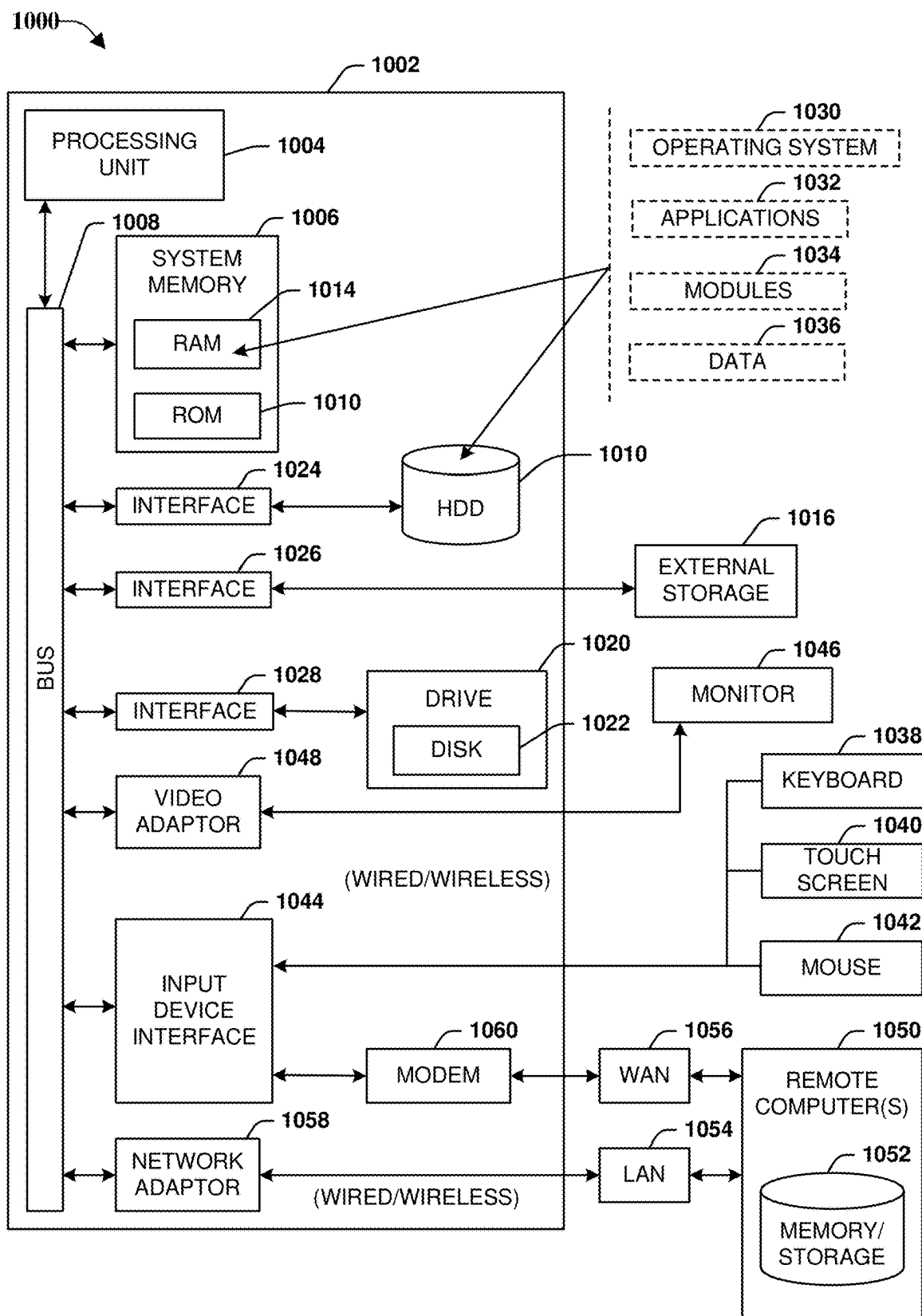


FIG. 10

**GENERATING AND UTILIZING
PROVISIONING TEMPLATES TO
PROVISION VOICE, VIDEO, AND DATA
COMMUNICATION SERVICES**

RELATED APPLICATION

[0001] The subject patent application is a continuation of U.S. patent application Ser. No. 18/307,771, filed Apr. 26, 2023, and entitled “Generating and Utilizing Provisioning Templates,” which is a continuation of U.S. patent application Ser. No. 17/355,396, filed Jun. 23, 2021, (now U.S. Pat. No. 11,671,316) and entitled “Generating and Utilizing Provisioning Templates to Provision Voice, Video, And Data Communication Services.” All sections of the aforementioned application(s) and/or patent(s) are incorporated herein by reference in their entirety.

TECHNICAL FIELD

[0002] The subject application is related to the implementation of networked computer systems and, for example, different approaches to provisioning services with computer systems.

BACKGROUND

[0003] Although handled by separate networks in existing systems, networks handling voice, video, and data communications have continued to be consolidated. Systems being consolidated often have applications built specifically to handle parts of the networks sought to be consolidated, e.g., separate applications for voice, video, and data communications. Problems can occur when the separate applications continue to be used for the consolidated network parts.

[0004] These problems are enhanced when services are sought to be provisioned within complex, consolidated systems. With the transition to next generation communications systems, and the system changes that these systems can require, frequent and significant changes to provisioning processes can be required. In some circumstances, even small changes to provisioning processes can require modifying large amounts of specialized computer code and the shutdown of network systems for long periods of time.

BRIEF DESCRIPTION OF THE DRAWINGS

[0005] The technology described herein is illustrated by way of example and not limited in the accompanying figures in which like reference numerals indicate similar elements and in which:

[0006] FIG. 1 is an architecture diagram of an example system that can facilitate generating and utilizing logical provisioning templates to provision voice, video, and data communication services, in accordance with one or more embodiments.

[0007] FIG. 2 illustrates different processes used by one or more embodiments to generate and utilize logical provisioning templates to provision voice, video, and data communication services, in accordance with one or more embodiments.

[0008] FIG. 3 is a layered architectural diagram of a non-limiting example system that can facilitate generating and utilizing logical provisioning templates to provision voice, video, and data communication services, in accordance with one or more embodiments.

[0009] FIG. 4 is an architecture diagram of an example system that continues the discussion of provisioning component, in accordance with one or more embodiments.

[0010] FIG. 5 is an architecture diagram of an example system that can illustrate example operations of dynamic execution environment, in accordance with one or more embodiments.

[0011] FIG. 6 is a diagram of a non-limiting example method that can facilitate generating and utilizing logical provisioning templates to provision voice, video, and data communication services, in accordance with one or more embodiments.

[0012] FIG. 7 depicts a system where one or more functions of operations support equipment 150 described above can be implemented to facilitate generating and utilizing logical provisioning templates to provision voice, video, and data communication services, in accordance with one or more embodiments.

[0013] FIG. 8 depicts an example non-transitory machine-readable medium that can tangibly store executable instructions that, when executed by a processor of a system, facilitate generating and utilizing logical provisioning templates to provision voice, video, and data communication services, in accordance with one or more embodiments described above.

[0014] FIG. 9 illustrates an example block diagram of an example mobile handset operable to engage in a system architecture that can facilitate processes described herein, in accordance with one or more embodiments.

[0015] FIG. 10 illustrates an example block diagram of an example computer operable to engage in a system architecture that can facilitate processes described herein, in accordance with one or more embodiments.

DETAILED DESCRIPTION

[0016] Aspects of the subject disclosure will now be described more fully hereinafter with reference to the accompanying drawings in which example components, graphs and selected operations are shown. In the following description, for purposes of explanation, numerous specific details are set forth in order to provide a thorough understanding of the various embodiments. For example, some embodiments described can facilitate generating and utilizing logical provisioning templates to provision voice, video, and data communication services. Different examples that describe these aspects are included with the description of FIGS. 1-10 below. It should be noted that the subject disclosure may be embodied in many different forms and should not be construed as limited to this example or other examples set forth herein.

[0017] In at least one implementation, one or more embodiments can be used to facilitate the implementation of parts of a large-scale wireless communication network that spans various geographic areas. According to this implementation, the one or more communication service provider networks can be or include the wireless communication network and/or various additional devices and components of the wireless communication network (e.g., additional network devices and cells, additional user equipment, network server devices, etc.). One or more embodiments can facilitate the placement of multiple antennas in a geographic area enabling coverage by networks that include, but are not limited to, communication service provider networks. In exemplary, non-limiting embodiments described herein,

simulated groups of antennas can include millimeter wave (mmWave) antennas of a base station of a cellular network, e.g., a fifth generation or other next generation radio access network (RAN). In certain implementations of types of mmWave antennas, large numbers of antennas are deployed in comparison with the number of other types of antennas that can be employed, e.g., up to and exceeding in some circumstances, one hundred antennas per square mile.

[0018] In some embodiments the non-limiting terms “signal propagation equipment” or simply “propagation equipment,” “radio network node” or simply “network node,” “radio network device,” “network device,” and access elements are used herein. These terms may be used interchangeably, and refer to any type of network node that can serve user equipment and/or be connected to other network node or network element or any radio node from where user equipment can receive a signal. Examples of radio network node include, but are not limited to, base stations (BS), multi-standard radio (MSR) nodes such as MSR BS, gNodeB, eNode B, network controllers, radio network controllers (RNC), base station controllers (BSC), relay, donor node controlling relay, base transceiver stations (BTS), access points (AP), transmission points, transmission nodes, remote radio units (RRU) (also termed radio units herein), remote radio heads (RRH), and nodes in distributed antenna system (DAS). Additional types of nodes are also discussed with embodiments below, e.g., donor node equipment and relay node equipment, an example use of these being in a network with an integrated access backhaul network topology.

[0019] In some embodiments, the non-limiting term user equipment (UE) is used. This term can refer to any type of wireless device that can communicate with a radio network node in a cellular or mobile communication system. Examples of UEs include, but are not limited to, a target device, device to device (D2D) user equipment, machine type user equipment, user equipment capable of machine to machine (M2M) communication, PDAs, tablets, mobile terminals, smart phones, laptop embedded equipped (LEE), laptop mounted equipment (LME), USB dongles, and other equipment that can have similar connectivity. Example UEs are described further with FIGS. 9 and 10 below. Some embodiments are described in particular for 5G new radio systems. The embodiments are however applicable to any radio access technology (RAT) or multi-RAT system where the UEs operate using multiple carriers, e.g., LTE.

[0020] The computer processing systems, computer-implemented methods, apparatus and/or computer program products described herein employ hardware and/or software to solve problems that are highly technical in nature (e.g., utilizing logical provisioning templates to deploy complex hardware and software configurations), that are not abstract and cannot be performed as a set of mental acts by a human. For example, a human, or even a plurality of humans, cannot efficiently deploy and configure network equipment in accordance with logical and physical models, with the same level of accuracy and/or efficiency as the various embodiments described herein.

[0021] Aspects of the subject disclosure will now be described more fully hereinafter with reference to the accompanying drawings in which example components, graphs and selected operations are shown. In the following description, for purposes of explanation, numerous specific details are set forth in order to provide a thorough under-

standing of the various embodiments. For example, some embodiments described can facilitate generating and utilizing logical provisioning templates to provision voice, video, and data communication services.

[0022] As discussed further below, one or more embodiments can address problems such as those described in the Background section above with inventive concepts that include different approaches to generating and utilizing logical provisioning templates to provision voice, video, and data communication services. Generally speaking, with logical provisioning templates and other approaches described herein, one or more embodiments can provide a dynamic provisioning platform that can facilitate flexible provisioning of services across a variety of new and legacy systems.

[0023] FIG. 1 is an architecture diagram of an example system **100** that can facilitate generating and utilizing logical provisioning templates to provision voice, video, and data communication services, in accordance with one or more embodiments. For purposes of brevity, description of like elements and/or processes employed in other embodiments is omitted.

[0024] As depicted, system **100** can include operations support equipment **150** receiving network provisioning request **135**, and communicatively coupled to network equipment **180**, service implementation equipment **175A-B**, via network **190**. In one or more embodiments, operations support equipment **150** can include computer executable components **120**, processor **160**, storage device **162**, memory **165**, and logical provisioning templates **196**. Computer executable components **120** can include request receiving component **122**, template identifying component **124**, dynamic execution environment **126**, provisioning component **128**, and other components described or suggested by different embodiments described herein that can improve the operation of system **100**.

[0025] It is noted that these components, as well as aspects of the embodiments of the subject disclosure depicted in this figure and various figures disclosed herein, are for illustration only, and as such, the architecture of such embodiments are not limited to the systems, devices, and/or components depicted therein. For example, in some embodiments, operations support equipment **150** can further comprise various computer and/or computing-based elements described herein with reference to operating environment **1000** and FIG. 10.

[0026] According to multiple embodiments, network **190** can comprise, but is not limited to, wired and wireless networks, including, but not limited to, a cellular network, a wide area network (WAN) (e.g., the Internet) or a local area network (LAN). For example, system **100** can communicate with one or more external systems, sources, and/or devices, for instance, computing devices (and vice versa) using virtually any desired wired or wireless technology, including but not limited to: wireless fidelity (Wi-Fi), global system for mobile communications (GSM), universal mobile telecommunications system (UMTS), worldwide interoperability for microwave access (WiMAX), enhanced general packet radio service (enhanced GPRS), third generation partnership project (3GPP) long term evolution (LTE), third generation partnership project 2 (3GPP2) ultra-mobile broadband (UMB), high speed packet access (HSPA), Zigbee and other 802.XX wireless technologies and/or legacy telecommunication technologies, BLUETOOTH®, Session Initiation Protocol (SIP), ZIGBEE®, RF4CE protocol,

WirelessHART protocol, 6LoWPAN (IPv6 over Low power Wireless Area Networks), Z-Wave, an ANT, an ultra-wide-band (UWB) standard protocol, and/or other proprietary and non-proprietary communication protocols.

[0027] According to multiple embodiments, storage device **162** can include, but is not limited to, RAM, ROM, EEPROM, flash memory or other memory technology, solid state drive (SSD) or other solid-state storage technology, Compact Disk Read Only Memory (CD ROM), digital video disk (DVD), blu-ray disk, or other optical disk storage, magnetic cassettes, magnetic tape, magnetic disk storage or other magnetic storage devices, or any other medium which can be used to store the desired information and which can be accessed by the computer.

[0028] According to multiple embodiments, processor **160** can comprise one or more processors and/or electronic circuitry that can implement one or more computer and/or machine readable, writable, and/or executable components and/or instructions that can be stored on memory **165**. For example, processor **160** can perform various operations that can be specified by such computer and/or machine readable, writable, and/or executable components and/or instructions including, but not limited to, logic, control, input/output (I/O), arithmetic, and/or the like. In some embodiments, processor **160** can comprise one or more components including, but not limited to, a central processing unit, a multi-core processor, a microprocessor, dual microprocessors, a micro-controller, a system on a chip (SOC), an array processor, a vector processor, and other types of processors. Further examples of processor **160** are described below with reference to processing unit **1004** of FIG. **10**. Such examples of processor **160** can be employed to implement any embodiments of the subject disclosure.

[0029] According to multiple embodiments, operations support equipment **150** can include memory **165**. In some embodiments, memory **165** can comprise volatile memory (e.g., random access memory (RAM), static RAM (SRAM), dynamic RAM (DRAM), etc.) and/or non-volatile memory (e.g., read only memory (ROM), programmable ROM (PROM), electrically programmable ROM (EPROM), electrically erasable programmable ROM (EEPROM), etc.) that can employ one or more memory architectures. Further examples of memory **165** are described below with reference to system memory **1006** and FIG. **10**. Such examples of memory **165** can be employed to implement any embodiments of the subject disclosure.

[0030] In some embodiments, memory **165** can store one or more non-transitory computer and/or machine readable, writable, and/or executable components and/or instructions **120** that, when respectively executed by processor **160**, can facilitate performance of operations defined by the executable component(s) and/or instruction(s). Generally, applications (e.g., computer executable components **120**) can include routines, programs, components, data structures, etc., that perform particular tasks or implement particular abstract data types. Moreover, those skilled in the art will appreciate that the methods described herein can be practiced with other system configurations, including single-processor or multiprocessor systems, minicomputers, main-frame computers, as well as personal computers, hand-held computing devices, microprocessor-based or programmable consumer electronics, and the like, each of which can be operatively coupled to one or more associated devices.

[0031] In one or more embodiments, computer executable components **120** can be used in connection with implementing one or more of the systems, devices, components, and/or computer-implemented operations shown and described in connection with FIG. **1** or other figures disclosed herein. For example, in one or more embodiments, computer executable components **120** can include instructions that, when executed by processor **160**, can facilitate performance of operations defining functional components, e.g., request receiving component **122**. As discussed with examples of one or more embodiments below, request receiving component **122** can, in accordance with one or more embodiments, facilitate receiving a first provisioning request for a network provisioning task. One having skill in the relevant art(s), given the description herein, will appreciate that the provisioning request can be applied to provisioning services from service implementation equipment **175A-B** in ways that can benefit from implementation using approaches described herein.

[0032] In another example, in one or more embodiments, computer executable components **120** can include instructions that, when executed by processor **160**, can facilitate performance of operations defining template identifying component **124**. As discussed below, template identifying component **124** can, in accordance with one or more embodiments, based on the network provisioning task, select a first logical provisioning template to facilitate a performance of the network provisioning task.

[0033] In additional embodiments, computer executable components **120** can include instructions that, when executed by processor **160**, can facilitate performance of operations defining dynamic execution environment **126**. As discussed below, dynamic execution environment **126** can, in accordance with one or more embodiments, based on the first logical provisioning template, identify a first programming object comprising logical commands for the performance of the network provisioning task.

[0034] In additional embodiments, computer executable components **120** can include instructions that, when executed by processor **160**, can facilitate performance of operations defining provisioning component **128**. As discussed below, provisioning component **128** can, in accordance with one or more embodiments, execute the logical commands of the first programming object to control (e.g., by dynamic execution environment **126**) service implementation equipment **175A-B** to perform the network provisioning task.

[0035] It is noted that the embodiments of the subject disclosure depicted in various figures disclosed herein are for illustration only, and as such, the architecture of such embodiments are not limited to the systems, devices, and/or components depicted therein. For example, in some embodiments, operations support equipment **150** can further comprise various computer and/or computing-based elements described herein with reference to operating environment **1000** and FIG. **10**. In one or more embodiments, such computer and/or computing-based elements can be used in connection with implementing one or more of the systems, devices, components, and/or computer-implemented operations shown and described in connection with FIG. **1** or other figures disclosed herein.

[0036] FIG. **2** illustrates different processes used by one or more embodiments to generate and utilize logical provisioning templates **196** to provision voice, video, and data com-

munication services, in accordance with one or more embodiments. As depicted, system 200 broadly includes equipment and components from provider network 255 and includes operations support system 250 (e.g., operations support equipment 150). Provider network 255 can include content sources 210 linked to common access network 202, common provider edge 204, and next generation provider core (P-Core) 206. Operations support equipment 150 can include provisioning component 128, with workflow engine 245 and business logic 275 applied to communications to parts of provider network 255, and data store 230 providing stored data to provisioning component 128.

[0037] In one or more embodiments, business logic 275 can be embodied by programming object 242, e.g., generated by provisioning component 128 based on network provisioning request 135 and logical provisioning template 196 selected by template identifying component 124. In an example implementation, programming object 242 can include logical commands to implement the network provisioning task, independent of commands used by the service implementation equipment 175A-B. Based at least on this approach, one or more embodiments can separate the business logic that guides provisioning tasks from the programming commands that implement the logic. One having skill in the relevant art(s), given this description of this approach, would appreciate that this separation can some of the benefits described and suggested by the descriptions herein.

[0038] Further, in one or more embodiments, programming object 242 can be considered as a reusable component, e.g., with rules, reference data, and instructions that are reused to perform the provisioning actions. In different embodiments, this use of business logic component can reduce the overhead associated with adding or changing the business logic used, e.g., once a component (e.g., functions, flows, scripts, reference data) is created/tested, it can be automatically picked up by the system for conversion (e.g., by dynamic execution environment 126) into equipment-specific instructions for service implementation equipment 175A, with no static data models that require code generation and building. Thus, in some implementations of one or more embodiments, creating, deploying, and changing the business logic that controls provisioning system services can require no separate installation procedure, no system restart, with one or more embodiments facilitating systems being almost-always running, with minimal downtime, e.g., termed by some embodiments as continuous integration and continuous delivery systems. Further characteristics of one or more embodiments can include benefits from application functions being built using a consistent architecture, e.g., with platform tooling remaining consistent across organizations. For example, one or more embodiments can utilize one common way of performing functions that include, but are not limited to, authentication, authorization, logging, load balancing, fault tolerance, and key performance indicator (KPI). In another example, applications deployed using one or more embodiments can be portable across on-premises and cloud platforms with minimal changes.

[0039] In another example, based on the separation of logical commands from implementation commands described above, one or more embodiments can use the same logical commands to control other service implementation equipment 175B to perform the network provisioning task concurrent with the first service implementation equipment 175A performing the network provisioning task, e.g., two

different implementation devices, with different commands required for use, can be operated concurrently using business logic 275 of programming object 242.

[0040] In additional or alternative embodiments, the executing of the logical commands of the programming object 242 can utilize application data from a data store 230. One having skill in the relevant art(s), given the description herein would appreciate that this use of application data can provide data-centric performance of network provisioning tasks by embodiments, e.g., in contrast to application-centric approaches of some legacy applications with isolated application data.

[0041] In additional or alternative embodiments, executing programming object 242 can control service implementation equipment 175A-B according to a workflow, e.g., by utilizing workflow engine 245.

[0042] FIG. 3 is a layered architectural diagram of a non-limiting example system 300 that can facilitate generating and utilizing logical provisioning templates 196 to provision voice, video, and data communication services, in accordance with one or more embodiments. For purposes of brevity, description of like elements and/or processes employed in other embodiments is omitted. As depicted, system 300 can include data store 230 with validation component 352, models 354, templates 356 (e.g., logical provisioning templates 196), metadata 358, and provisioning application data 360.

[0043] As depicted, data store 230 provides examples of different kinds of data that can be moved from isolated service implementation equipment 175A-B, to the integrated, data-centric approaches of one or more embodiments. For example, with isolated, application-centric approaches to provisioning, uniform validation of application data can be difficult, e.g., with each of service implementation equipment 175A-B having to independently validate data using the same rules. In contrast, one or more embodiments can, with validation component 352, control service implementation equipment 175A-B based on the same validation criteria, e.g., from sources that can include, but are not limited to, provisioning application data 360 and logical provisioning templates 196.

[0044] An example format that can be used by one or more embodiments for network provisioning request is JavaScript Object Notation (JSON), but this example is non-limiting, and other formatting can be used without deviating from the disclosure herein. Further to this example, in one or more embodiments, validation component 352 can include, but is not limited to, a JSON schema validator for use validating JSON formatted information.

[0045] FIG. 4 is an architecture diagram of an example system 400 that continues the discussion of provisioning component 128, in accordance with one or more embodiments. For purposes of brevity, description of like elements and/or processes employed in other embodiments is omitted. As depicted, system 400 can include provisioning component 128 communicatively coupled to data store 230, with provisioning component 128 including script generating engine 450, and logical provisioning template 196 examples: passive optical network (PON) design template 440A, ethernet network design template 440B, multiprotocol label switching (MPLS) network design template, MPLS network design template 440C, and IP network design template 440D.

[0046] In an example of provisioning by one or more embodiments, network provisioning request 135 can include provisioning of broadband internet service node equipment of a broadband network. Based on this task, template identifying component 124 can select one or more of passive optical network (PON) design template 440A, MPLS network design template 440C, and IP network design template 440D. These design templates can be used to generate programming object 242 with business logic 275 from a broad variety of implementation steps, model approaches, workflows, validation rules, and application data associated with this task of provisioning broadband internet service.

[0047] To facilitate the completion of this task, based on the selected templates, data store 230, and other information about available network resources, one or more embodiments can identify specific broadband network gateway equipment that can be configured and activated for the task. Based on this identified service implementation equipment 175A, one or more embodiments can select specific configuration and activation commands for the broadband network gateway equipment (e.g., including administrative information such as sufficient credentials) to implement the logical processes of programming object 242.

[0048] In another example of provisioning by one or more embodiments, network provisioning request 135 can include provisioning of a label-switching network connection for an enterprise customer. Based on this task, template identifying component 124 can select multiprotocol label switching (MPLS) network design template, with this design template being used to generate another instance of programming object 242 with business logic 275 associated with this task of MPLS network implementation.

[0049] To facilitate the completion of this task, based on the selected template, data store 230, and other information about available network resources, one or more embodiments can identify specific service implementation equipment 175B (e.g., packet-forwarding equipment) that can be configured and activated for the task. Based on this identified service implementation equipment 175B, one or more embodiments can select specific configuration and activation commands for the packet-forwarding equipment.

[0050] In additional or alternative embodiments, the logical commands of programming object 242 can include non-compiled computer script generated by script generating engine 450 to control the service implementation equipment. One benefit that can accrue from this approach is the capability, by some embodiments, to change the processes and parameters of different provisioning tasks with a reduction of elimination of recompiling, redeploying, and rebooting network equipment.

[0051] FIG. 5 is an architecture diagram of an example system 500 that can illustrate example operations of dynamic execution environment 126, in accordance with one or more embodiments. For purposes of brevity, description of like elements and/or processes employed in other embodiments is omitted. In one or more embodiments, computer-executable instructions 220 can include additional components discussed below, e.g., to work with the operation of provisioning component 128 described with FIGS. 3-4 above.

[0052] As discussed above, provisioning component 128 can use programming object 242 to generate script for conversion by dynamic execution environment 126 into

commands specific to service implementation equipment 175A-B, e.g., with the script being interpreted by script interpreting engine 430.

[0053] FIG. 6 is a diagram of a non-limiting example method 600 that can facilitate generating and utilizing logical provisioning templates 196 to provision voice, video, and data communication services, in accordance with one or more embodiments. For purposes of brevity, description of like elements and/or processes employed in other embodiments is omitted.

[0054] It should be noted that, in one or more embodiments, system 100 and other embodiments described herein can employ hardware and/or software to solve problems that are highly technical in nature, including modeling complex hardware and software deployments. One having skill in the relevant art(s), given the disclosure herein, would appreciate that the technical problems that can be solved by one or more embodiments described herein are not abstract and cannot be performed as a set of mental acts by a human, e.g., in some circumstances updating complex models of network equipment based on detected changes to systems, e.g., based on change request 410.

[0055] Further, in certain embodiments, some of the processes performed can be performed by one or more specialized computers (e.g., one or more specialized processing units, a specialized computer such as tomography and reconstruction, statistical estimation, specialized routing analysis, and so on) for carrying out defined tasks related to timing the performance of change procedures for systems where system functions are implemented with different redundant safeguards. System 100 and/or components of the system can be employed to use method 600 and other embodiments to solve new problems that arise through advancements in technologies mentioned above, computer architecture, and/or the like.

[0056] At 602, method 600 can include receiving a first provisioning request for a network provisioning task. At 604, method 600 can include, based on the network provisioning task, selecting a first logical provisioning template to facilitate a performance of the network provisioning task. At 606, method 600 can include based on the first logical provisioning template, identifying a first programming object comprising logical commands for the performance of the network provisioning task. At 608, method 600 can include executing the logical commands of the first programming object to control first service implementation equipment to perform the network provisioning task.

[0057] FIG. 7 depicts a system 700 where one or more functions of operations support equipment 150 described above can be implemented to facilitate generating and utilizing logical provisioning templates 196 to provision voice, video, and data communication services, in accordance with one or more embodiments. For purposes of brevity, description of like elements and/or processes employed in other embodiments is omitted.

[0058] In one or more embodiments, operations support equipment 150 can be implemented in a software platform that includes several interconnected components. As depicted, system 700 can include request receiving component 122, template identifying component 124, dynamic execution environment 126, provisioning component 128, and other components described or suggested by different embodiments described herein.

[0059] In an example, component **702** can include the functions of request receiving component **122**, supported by the other layers of system **700**. For example, component **702** can receive a first provisioning request for a network provisioning task. In another example, component **704** can include the functions of template identifying component **124**, supported by the other layers of system **700**. For example, component **704** can based on the network provisioning task, select a first logical provisioning template to facilitate a performance of the network provisioning task.

[0060] In yet another example, component **706** can include the functions of dynamic execution environment **126**, supported by the other layers of system **700**. For example, component **706** can, based on the first logical provisioning template, identify a first programming object comprising logical commands for the performance of the network provisioning task. In yet another example, component **708** can include the functions of dynamic execution environment **126**, supported by the other layers of system **700**. For example, component **708** can execute the logical commands of the first programming object to control first service implementation equipment to perform the network provisioning task.

[0061] FIG. **8** depicts an example **800** non-transitory machine-readable medium **810** that can tangibly store executable instructions that, when executed by a processor of a system, facilitate generating and utilizing logical provisioning templates **196** to provision voice, video, and data communication services, in accordance with one or more embodiments described above. For purposes of brevity, description of like elements and/or processes employed in other embodiments is omitted. As depicted, non-transitory machine-readable medium **810** includes executable instructions that can facilitate performance of operations **802-806**.

[0062] In one or more embodiments, the operations can include operation **802** to receive a first provisioning request for a network provisioning task. In one or more embodiments, the operations can include operation **804** to, based on the network provisioning task, select a first logical provisioning template to facilitate a performance of the network provisioning task. In one or more embodiments, the operations can include operation **806** to, based on the first logical provisioning template, identify a first programming object comprising logical commands for the performance of the network provisioning task. In one or more embodiments, the operations can include operation **808** to execute the logical commands of the first programming object to control first service implementation equipment to perform the network provisioning task.

[0063] FIG. **9** illustrates an example block diagram of an example mobile handset **900** operable to engage in a system architecture that facilitates wireless communications according to one or more embodiments described herein. Although a mobile handset is illustrated herein, it will be understood that other devices can be a mobile device, and that the mobile handset is merely illustrated to provide context for the embodiments of the various embodiments described herein. The following discussion is intended to provide a brief, general description of an example of a suitable environment in which the various embodiments can be implemented. While the description includes a general context of computer-executable instructions embodied on a machine-readable storage medium, those skilled in the art will recognize that the embodiments also can be implemented in

combination with other program modules and/or as a combination of hardware and software.

[0064] Generally, applications (e.g., program modules) can include routines, programs, components, data structures, etc., that perform particular tasks or implement particular abstract data types. Moreover, those skilled in the art will appreciate that the methods described herein can be practiced with other system configurations, including single-processor or multiprocessor systems, minicomputers, main-frame computers, as well as personal computers, hand-held computing devices, microprocessor-based or programmable consumer electronics, and the like, each of which can be operatively coupled to one or more associated devices

[0065] A computing device can typically include a variety of machine-readable media. Machine-readable media can be any available media that can be accessed by the computer and includes both volatile and non-volatile media, removable and non-removable media. By way of example and not limitation, computer-readable media can comprise computer storage media and communication media. Computer storage media can include volatile and/or non-volatile media, removable and/or non-removable media implemented in any method or technology for storage of information, such as computer-readable instructions, data structures, program modules, or other data. Computer storage media can include, but is not limited to, RAM, ROM, EEPROM, flash memory or other memory technology, solid state drive (SSD) or other solid-state storage technology, Compact Disk Read Only Memory (CD ROM), digital video disk (DVD), Blu-ray disk, or other optical disk storage, magnetic cassettes, magnetic tape, magnetic disk storage or other magnetic storage devices, or any other medium which can be used to store the desired information and which can be accessed by the computer. In this regard, the terms “tangible” or “non-transitory” herein as applied to storage, memory or computer-readable media, are to be understood to exclude only propagating transitory signals per se as modifiers and do not relinquish rights to all standard storage, memory or computer-readable media that are not only propagating transitory signals per se.

[0066] Communication media typically embodies computer-readable instructions, data structures, program modules, or other data in a modulated data signal such as a carrier wave or other transport mechanism, and includes any information delivery media. The term “modulated data signal” means a signal that has one or more of its characteristics set or changed in such a manner as to encode information in the signal. By way of example, and not limitation, communication media includes wired media such as a wired network or direct-wired connection, and wireless media such as acoustic, RF, infrared and other wireless media. Combinations of the any of the above should also be included within the scope of computer-readable media

[0067] The handset includes a processor **902** for controlling and processing all onboard operations and functions. A memory **904** interfaces to the processor **902** for storage of data and one or more applications **906** (e.g., a video player software, user feedback component software, etc.). Other applications can include voice recognition of predetermined voice commands that facilitate initiation of the user feedback signals. The applications **906** can be stored in the memory **904** and/or in a firmware **908**, and executed by the processor **902** from either or both the memory **904** and/or the firmware **908**. The firmware **908** can also store startup code

for execution in initializing the handset **900**. A communications component **910** interfaces to the processor **902** to facilitate wired/wireless communication with external systems, e.g., cellular networks, VOIP networks, and so on. Here, the communications component **910** can also include a suitable cellular transceiver **913** (e.g., a GSM transceiver) and/or an unlicensed transceiver **913** (e.g., Wi-Fi, WiMax) for corresponding signal communications. The handset **900** can be a device such as a cellular telephone, a PDA with mobile communications capabilities, and messaging-centric devices. The communications component **910** also facilitates communications reception from terrestrial radio networks (e.g., broadcast), digital satellite radio networks, and Internet-based radio services networks.

[0068] The handset **900** includes a display **912** for displaying text, images, video, telephony functions (e.g., a Caller ID function), setup functions, and for user input. For example, the display **912** can also be referred to as a “screen” that can accommodate the presentation of multimedia content (e.g., music metadata, messages, wallpaper, graphics, etc.). The display **912** can also display videos and can facilitate the generation, editing and sharing of video quotes. A serial I/O interface **914** is provided in communication with the processor **902** to facilitate wired and/or wireless serial communications (e.g., USB, and/or IEEE 1294) through a hardwire connection, and other serial input devices (e.g., a keyboard, keypad, and mouse). This supports updating and troubleshooting the handset **900**, for example. Audio capabilities are provided with an audio I/O component **916**, which can include a speaker for the output of audio signals related to, for example, indication that the user pressed the proper key or key combination to initiate the user feedback signal. The audio I/O component **916** also facilitates the input of audio signals through a microphone to record data and/or telephony voice data, and for inputting voice signals for telephone conversations.

[0069] The handset **900** can include a slot interface **918** for accommodating a SIC (Subscriber Identity Component) in the form factor of a card Subscriber Identity Module (SIM) or universal SIM **920**, and interfacing the SIM card **920** with the processor **902**. However, it is to be appreciated that the SIM card **920** can be manufactured into the handset **900**, and updated by downloading data and software.

[0070] The handset **900** can process IP data traffic through the communications component **910** to accommodate IP traffic from an IP network such as, for example, the Internet, a corporate intranet, a home network, a person area network, etc., through an ISP or broadband cable provider. Thus, VoIP traffic can be utilized by the handset **900** and IP-based multimedia content can be received in either an encoded or a decoded format.

[0071] A video processing component **922** (e.g., a camera) can be provided for decoding encoded multimedia content. The video processing component **922** can aid in facilitating the generation, editing, and sharing of video quotes. The handset **900** also includes a power source **924** in the form of batteries and/or an AC power subsystem, which power source **924** can interface to an external power system or charging equipment (not shown) by a power I/O component **926**.

[0072] The handset **900** can also include a video component **930** for processing video content received and for recording and transmitting video content. For example, the video component **930** can facilitate the generation, editing

and sharing of video quotes. A location tracking component **932** facilitates geographically locating the handset **900**. As described hereinabove, this can occur when the user initiates the feedback signal automatically or manually. A user input component **934** facilitates the user initiating the quality feedback signal. The user input component **934** can also facilitate the generation, editing and sharing of video quotes. The user input component **934** can include such conventional input device technologies such as a keypad, keyboard, mouse, stylus pen, and/or touch screen, for example.

[0073] Referring again to the applications **906**, a hysteresis component **936** facilitates the analysis and processing of hysteresis data, which is utilized to determine when to associate with the access point. A software trigger component **938** can be provided that facilitates triggering of the hysteresis component **936** when the Wi-Fi transceiver **913** detects the beacon of the access point. A SIP client **940** enables the handset **900** to support SIP protocols and register the subscriber with the SIP registrar server. The applications **906** can also include a client **946** that provides at least the capability of discovery, play and store of multimedia content, for example, music.

[0074] The handset **900**, as indicated above related to the communications component **910**, includes an indoor network radio transceiver **913** (e.g., Wi-Fi transceiver). This function supports the indoor radio link, such as IEEE 802.11, for the dual-mode GSM handset **900**. The handset **900** can accommodate at least satellite radio services through a handset that can combine wireless voice and digital radio chipsets into a single handheld device.

[0075] Network **190** can employ various cellular systems, technologies, and modulation schemes to facilitate wireless radio communications between devices. While example embodiments include use of 5G new radio (NR) systems, one or more embodiments discussed herein can be applicable to any radio access technology (RAT) or multi-RAT system, including where different user equipment operate using multiple carriers, e.g., LTE FDD/TDD, GSM/GERAN, CDMA2000, etc. For example, wireless communication system **200** can operate in accordance with global system for mobile communications (GSM), universal mobile telecommunications service (UMTS), long term evolution (LTE), LTE frequency division duplexing (LTE FDD, LTE time division duplexing (TDD), high speed packet access (HSPA), code division multiple access (CDMA), wideband CDMA (WCMDA), CDMA2000, time division multiple access (TDMA), frequency division multiple access (FDMA), multi-carrier code division multiple access (MC-CDMA), single-carrier code division multiple access (SC-CDMA), single-carrier FDMA (SC-FDMA), orthogonal frequency division multiplexing (OFDM), discrete Fourier transform spread OFDM (DFT-spread OFDM) single carrier FDMA (SC-FDMA), Filter bank based multi-carrier (FBMC), zero tail DFT-spread-OFDM (ZT DFT-s-OFDM), generalized frequency division multiplexing (GFDM), fixed mobile convergence (FMC), universal fixed mobile convergence (UFMC), unique word OFDM (UW-OFDM), unique word DFT-spread OFDM (UW DFT-Spread-OFDM), cyclic prefix OFDM CP-OFDM, resource-block-filtered OFDM, Wi Fi, WLAN, WiMax, and the like. However, various features and functionalities of system **100** are particularly described wherein the devices of system **100** are configured to communicate wireless signals using one or more multi carrier modulation schemes, wherein data symbols can be

transmitted simultaneously over multiple frequency subcarriers (e.g., OFDM, CP-OFDM, DFT-spread OFDM, UMC, FMBC, etc.). The embodiments are applicable to single carrier as well as to multicarrier (MC) or carrier aggregation (CA) operation of the user equipment. The term carrier aggregation (CA) is also called (e.g., interchangeably called) “multi-carrier system”, “multi-cell operation”, “multi-carrier operation”, “multi-carrier” transmission and/or reception. Note that some embodiments are also applicable for Multi RAB (radio bearers) on some carriers (that is data plus speech is simultaneously scheduled).

[0076] Various embodiments described herein can be configured to provide and employ 5G wireless networking features and functionalities. With 5G networks that may use waveforms that split the bandwidth into several sub bands, different types of services can be accommodated in different sub bands with the most suitable waveform and numerology, leading to improved spectrum utilization for 5G networks. Notwithstanding, in the mmWave spectrum, the millimeter waves have shorter wavelengths relative to other communications waves, whereby mm Wave signals can experience severe path loss, penetration loss, and fading. However, the shorter wavelength at mmWave frequencies also allows more antennas to be packed in the same physical dimension, which allows for large-scale spatial multiplexing and highly directional beamforming.

[0077] FIG. 10 provides additional context for various embodiments described herein, intended to provide a brief, general description of a suitable operating environment 1000 in which the various embodiments of the embodiment described herein can be implemented. While the embodiments have been described above in the general context of computer-executable instructions that can run on one or more computers, those skilled in the art will recognize that the embodiments can be also implemented in combination with other program modules and/or as a combination of hardware and software.

[0078] Generally, program modules include routines, programs, components, data structures, etc., that perform particular tasks or implement particular abstract data types. Moreover, those skilled in the art will appreciate that the various methods can be practiced with other computer system configurations, including single-processor or multi-processor computer systems, minicomputers, mainframe computers, Internet of Things (IoT) devices, distributed computing systems, as well as personal computers, handheld computing devices, microprocessor-based or programmable consumer electronics, and the like, each of which can be operatively coupled to one or more associated devices.

[0079] The illustrated embodiments of the embodiments herein can be also practiced in distributed computing environments where certain tasks are performed by remote processing devices that are linked through a communications network. In a distributed computing environment, program modules can be located in both local and remote memory storage devices.

[0080] Computing devices typically include a variety of media, which can include computer-readable storage media, machine-readable storage media, and/or communications media, which two terms are used herein differently from one another as follows. Computer-readable storage media or machine-readable storage media can be any available storage media that can be accessed by the computer and includes both volatile and nonvolatile media, removable and non-

removable media. By way of example, and not limitation, computer-readable storage media or machine-readable storage media can be implemented in connection with any method or technology for storage of information such as computer-readable or machine-readable instructions, program modules, structured data or unstructured data.

[0081] Computer-readable storage media can include, but are not limited to, random access memory (RAM), read only memory (ROM), electrically erasable programmable read only memory (EEPROM), flash memory or other memory technology, compact disk read only memory (CD ROM), digital versatile disk (DVD), Blu-ray disc (BD) or other optical disk storage, magnetic cassettes, magnetic tape, magnetic disk storage or other magnetic storage devices, solid state drives or other solid state storage devices, or other tangible and/or non-transitory media which can be used to store desired information. In this regard, the terms “tangible” or “non-transitory” herein as applied to storage, memory or computer-readable media, are to be understood to exclude only propagating transitory signals per se as modifiers and do not relinquish rights to all standard storage, memory or computer-readable media that are not only propagating transitory signals per se.

[0082] Computer-readable storage media can be accessed by one or more local or remote computing devices, e.g., via access requests, queries or other data retrieval protocols, for a variety of operations with respect to the information stored by the medium.

[0083] Communications media typically embody computer-readable instructions, data structures, program modules or other structured or unstructured data in a data signal such as a modulated data signal, e.g., a carrier wave or other transport mechanism, and includes any information delivery or transport media. The term “modulated data signal” or signals refers to a signal that has one or more of its characteristics set or changed in such a manner as to encode information in one or more signals. By way of example, and not limitation, communication media include wired media, such as a wired network or direct-wired connection, and wireless media such as acoustic, RF, infrared and other wireless media.

[0084] FIG. 10 depicts an example operating environment 1000 for implementing various embodiments of the aspects described herein includes a computer 1002, the computer 1002 including a processing unit 1004, a system memory 1006 and a system bus 1008. The system bus 1008 couples system components including, but not limited to, the system memory 1006 to the processing unit 1004. The processing unit 1004 can be any of various commercially available processors. Dual microprocessors and other multi processor architectures can also be employed as the processing unit 1004.

[0085] The system bus 1008 can be any of several types of bus structure that can further interconnect to a memory bus (with or without a memory controller), a peripheral bus, and a local bus using any of a variety of commercially available bus architectures. The system memory 1006 includes ROM 1014 and RAM 1012. A basic input/output system (BIOS) can be stored in a non-volatile memory such as ROM, erasable programmable read only memory (EPROM), EEPROM, which BIOS contains the basic routines that help to transfer information between elements within the com-

puter **1002**, such as during startup. The RAM **1012** can also include a high-speed RAM such as static RAM for caching data.

[0086] The computer **1002** further includes an internal hard disk drive (HDD) **1014** (e.g., EIDE, SATA), one or more external storage devices **1016** (e.g., a magnetic floppy disk drive (FDD) **1016**, a memory stick or flash drive reader, a memory card reader, etc.) and a drive **1020**, e.g., such as a solid state drive, an optical disk drive, which can read or write from a disk **1022**, such as a CD-ROM disc, a DVD, a BD, etc. Alternatively, where a solid-state drive is involved, disk **1022** would not be included, unless separate. While the internal HDD **1014** is illustrated as located within the computer **1002**, the internal HDD **1014** can also be configured for external use in a suitable chassis (not shown). Additionally, while not shown in environment **1000**, a solid state drive (SSD) could be used in addition to, or in place of, an HDD **1014**. The HDD **1014**, external storage device(s) **1016** and drive **1020** can be connected to the system bus **1008** by an HDD interface **1024**, an external storage interface **1026** and a drive interface **1028**, respectively. The interface **1024** for external drive implementations can include at least one or both of Universal Serial Bus (USB), ROM base address (RBA), and Institute of Electrical and Electronics Engineers (IEEE) 1394 interface technologies. Other external drive connection technologies are within contemplation of the embodiments described herein.

[0087] The drives and their associated computer-readable storage media provide nonvolatile storage of data, data structures, computer-executable instructions, and so forth. For the computer **1002**, the drives and storage media accommodate the storage of any data in a suitable digital format. Although the description of computer-readable storage media above refers to respective types of storage devices, it should be appreciated by those skilled in the art that other types of storage media which are readable by a computer, whether presently existing or developed in the future, could also be used in the example operating environment, and further, that any such storage media can contain computer-executable instructions for performing the methods described herein.

[0088] A number of program modules can be stored in the drives and RAM **1012**, including an operating system **1030**, one or more application programs **1032**, other program modules **1034** and program data **1036**. All or portions of the operating system, applications, modules, and/or data can also be cached in the RAM **1012**. The systems and methods described herein can be implemented utilizing various commercially available operating systems or combinations of operating systems.

[0089] Computer **1002** can optionally comprise emulation technologies. For example, a hypervisor (not shown) or other intermediary can emulate a hardware environment for operating system **1030**, and the emulated hardware can optionally be different from the hardware illustrated in FIG. **10**. In such an embodiment, operating system **1030** can comprise one virtual machine (VM) of multiple VMs hosted at computer **1002**. Furthermore, operating system **1030** can provide runtime environments, such as the Java runtime environment or the .NET framework, for applications **1032**. Runtime environments are consistent execution environments that allow applications **1032** to run on any operating system that includes the runtime environment. Similarly, operating system **1030** can support containers, and applica-

tions **1032** can be in the form of containers, which are lightweight, standalone, executable packages of software that include, e.g., code, runtime, system tools, system libraries and settings for an application.

[0090] Further, computer **1002** can be enabled with a security module, such as a trusted processing module (TPM). For instance, with a TPM, boot components hash next in time boot components, and wait for a match of results to secured values, before loading a next boot component. This process can take place at any layer in the code execution stack of computer **1002**, e.g., applied at the application execution level or at the operating system (OS) kernel level, thereby enabling security at any level of code execution.

[0091] A user can enter commands and information into the computer **1002** through one or more wired/wireless input devices, e.g., a keyboard **1038**, a touch screen **1040**, and a pointing device, such as a mouse **1042**. Other input devices (not shown) can include a microphone, an infrared (IR) remote control, a radio frequency (RF) remote control, or other remote control, a joystick, a virtual reality controller and/or virtual reality headset, a game pad, a stylus pen, an image input device, e.g., camera(s), a gesture sensor input device, a vision movement sensor input device, an emotion or facial detection device, a biometric input device, e.g., fingerprint or iris scanner, or the like. These and other input devices are often connected to the processing unit **1004** through an input device interface **1044** that can be coupled to the system bus **1008**, but can be connected by other interfaces, such as a parallel port, an IEEE 1394 serial port, a game port, a USB port, an IR interface, a BLUETOOTH- π /E interface, etc.

[0092] A monitor **1046** or other type of display device can be also connected to the system bus **1008** via an interface, such as a video adapter **1048**. In addition to the monitor **1046**, a computer typically includes other peripheral output devices (not shown), such as speakers, printers, etc.

[0093] The computer **1002** can operate in a networked environment using logical connections via wired and/or wireless communications to one or more remote computers, such as a remote computer(s) **1050**. The remote computer(s) **1050** can be a workstation, a server computer, a router, a personal computer, portable computer, microprocessor-based entertainment appliance, a peer device or other common network node, and typically includes many or all of the elements described relative to the computer **1002**, although, for purposes of brevity, only a memory/storage device **1052** is illustrated. The logical connections depicted include wired/wireless connectivity to a local area network (LAN) **1054** and/or larger networks, e.g., a wide area network (WAN) **1056**. Such LAN and WAN networking environments are commonplace in offices and companies, and facilitate enterprise-wide computer networks, such as intranets, all of which can connect to a global communications network, e.g., the Internet.

[0094] When used in a LAN networking environment, the computer **1002** can be connected to the local network **1054** through a wired and/or wireless communication network interface or adapter **1058**. The adapter **1058** can facilitate wired or wireless communication to the LAN **1054**, which can also include a wireless access point (AP) disposed thereon for communicating with the adapter **1058** in a wireless mode.

[0095] When used in a WAN networking environment, the computer **1002** can include a modem **1060** or can be

connected to a communications server on the WAN **1056** via other means for establishing communications over the WAN **1056**, such as by way of the Internet. The modem **1060**, which can be internal or external and a wired or wireless device, can be connected to the system bus **1008** via the input device interface **1044**. In a networked environment, program modules depicted relative to the computer **1002** or portions thereof, can be stored in the remote memory/storage device **1052**. It will be appreciated that the network connections shown are examples and other means of establishing a communications link between the computers can be used.

[0096] When used in either a LAN or WAN networking environment, the computer **1002** can access cloud storage systems or other network-based storage systems in addition to, or in place of, external storage devices **1016** as described above, such as but not limited to a network virtual machine providing one or more aspects of storage or processing of information. Generally, a connection between the computer **1002** and a cloud storage system can be established over a LAN **1054** or WAN **1056** e.g., by the adapter **1058** or modem **1060**, respectively. Upon connecting the computer **1002** to an associated cloud storage system, the external storage interface **1026** can, with the aid of the adapter **1058** and/or modem **1060**, manage storage provided by the cloud storage system as it would other types of external storage. For instance, the external storage interface **1026** can be configured to provide access to cloud storage sources as if those sources were physically connected to the computer **1002**.

[0097] The computer **1002** can be operable to communicate with any wireless devices or entities operatively disposed in wireless communication, e.g., a printer, scanner, desktop and/or portable computer, portable data assistant, communications satellite, any piece of equipment or location associated with a wirelessly detectable tag (e.g., a kiosk, news stand, store shelf, etc.), and telephone. This can include Wireless Fidelity (Wi-Fi) and BLUETOOTH™/E wireless technologies. Thus, the communication can be a predefined structure as with a conventional network or simply an ad hoc communication between at least two devices.

[0098] The above description of illustrated embodiments of the subject disclosure, including what is described in the Abstract, is not intended to be exhaustive or to limit the disclosed embodiments to the precise forms disclosed. While specific embodiments and examples are described herein for illustrative purposes, various modifications are possible that are considered within the scope of such embodiments and examples, as those skilled in the relevant art can recognize.

[0099] In this regard, while the disclosed subject matter has been described in connection with various embodiments and corresponding Figures, where applicable, it is to be understood that other similar embodiments can be used or modifications and additions can be made to the described embodiments for performing the same, similar, alternative, or substitute function of the disclosed subject matter without deviating therefrom. Therefore, the disclosed subject matter should not be limited to any single embodiment described herein, but rather should be construed in breadth and scope in accordance with the appended claims below.

[0100] Further to the description above, as it employed in the subject specification, the term “processor” can refer to

substantially any computing processing unit or device comprising, but not limited to comprising, single-core processors; single-processors with software multithread execution capability; multi-core processors; multi-core processors with software multithread execution capability; multi-core processors with hardware multithread technology; parallel platforms; and parallel platforms with distributed shared memory. Additionally, a processor can refer to an integrated circuit, an application specific integrated circuit (ASIC), a digital signal processor (DSP), a field programmable gate array (FPGA), a programmable logic controller (PLC), a complex programmable logic device (CPLD), a discrete gate or transistor logic, discrete hardware components, or any combination thereof designed to perform the functions described herein. Processors can exploit nano-scale architectures such as, but not limited to, molecular and quantum-dot based transistors, switches and gates, in order to optimize space usage or enhance performance of user equipment. A processor may also be implemented as a combination of computing processing units.

[0101] In the subject specification, terms such as “store,” “storage,” “data store,” “data storage,” “database,” and substantially any other information storage component relevant to operation and functionality of a component, refer to “memory components,” or entities embodied in a “memory” or components comprising the memory. It will be appreciated that the memory components described herein can be either volatile memory or nonvolatile memory, or can include both volatile and nonvolatile memory.

[0102] As used in this application, the terms “component,” “system,” “platform,” “layer,” “selector,” “interface,” and the like are intended to refer to a computer-related entity or an entity related to an operational apparatus with one or more specific functionalities, wherein the entity can be either hardware, a combination of hardware and software, software, or software in execution. As an example, a component may be, but is not limited to being, a process running on a processor, a processor, an object, an executable, a thread of execution, a program, and/or a computer. By way of illustration and not limitation, both an application running on a server and the server can be a component. One or more components may reside within a process and/or thread of execution and a component may be localized on one computer and/or distributed between two or more computers. In addition, these components can execute from various computer readable media, device readable storage devices, or machine readable media having various data structures stored thereon. The components may communicate via local and/or remote processes such as in accordance with a signal having one or more data packets (e.g., data from one component interacting with another component in a local system, distributed system, and/or across a network such as the Internet with other systems via the signal). As another example, a component can be an apparatus with specific functionality provided by mechanical parts operated by electric or electronic circuitry, which is operated by a software or firmware application executed by a processor, wherein the processor can be internal or external to the apparatus and executes at least a part of the software or firmware application. As yet another example, a component can be an apparatus that provides specific functionality through electronic components without mechanical parts, the electronic components can include a processor therein to

execute software or firmware that confers at least in part the functionality of the electronic components.

[0103] In addition, the term “or” is intended to mean an inclusive “or” rather than an exclusive “or.” That is, unless specified otherwise, or clear from context, “X employs A or B” is intended to mean any of the natural inclusive permutations. That is, if X employs A; X employs B; or X employs both A and B, then “X employs A or B” is satisfied under any of the foregoing instances. Moreover, articles “a” and “an” as used in the subject specification and annexed drawings should generally be construed to mean “one or more” unless specified otherwise or clear from context to be directed to a singular form.

[0104] Additionally, the terms “core-network”, “core”, “core carrier network”, “carrier-side”, or similar terms can refer to components of a telecommunications network that typically provides some or all of aggregation, authentication, call control and switching, charging, service invocation, or gateways. Aggregation can refer to the highest level of aggregation in a service provider network wherein the next level in the hierarchy under the core nodes is the distribution networks and then the edge networks. User equipment do not normally connect directly to the core networks of a large service provider but can be routed to the core by way of a switch or radio area network. Authentication can refer to determinations regarding whether the user requesting a service from the telecom network is authorized to do so within this network or not. Call control and switching can refer to determinations related to the future course of a call stream across carrier equipment based on the call signal processing. Charging can be related to the collation and processing of charging data generated by various network nodes. Two common types of charging mechanisms found in present day networks can be prepaid charging and postpaid charging. Service invocation can occur based on some explicit action (e.g., call transfer) or implicitly (e.g., call waiting). It is to be noted that service “execution” may or may not be a core network functionality as third party network/nodes may take part in actual service execution. A gateway can be present in the core network to access other networks. Gateway functionality can be dependent on the type of the interface with another network.

[0105] Furthermore, the terms “user,” “subscriber,” “customer,” “consumer,” “prosumer,” “agent,” and the like are employed interchangeably throughout the subject specification, unless context warrants particular distinction(s) among the terms. It should be appreciated that such terms can refer to human entities or automated components (e.g., supported through artificial intelligence, as through a capacity to make inferences based on complex mathematical formalisms), that can provide simulated vision, sound recognition and so forth.

[0106] Aspects, features, or advantages of the subject matter can be exploited in substantially any, or any, wired, broadcast, wireless telecommunication, radio technology or network, or combinations thereof. Non-limiting examples of such technologies or networks include Geocast technology; broadcast technologies (e.g., sub-Hz, ELF, VLF, LF, MF, HF, VHF, UHF, SHF, THz broadcasts, etc.); Ethernet; X.25; powerline-type networking (e.g., PowerLine AV Ethernet, etc.); femto-cell technology; Wi-Fi; Worldwide Interoperability for Microwave Access (WiMAX); Enhanced General Packet Radio Service (Enhanced GPRS); Third Generation Partnership Project (3GPP or 3G) Long Term Evolution

(LTE) or 5G; 3GPP Universal Mobile Telecommunications System (UMTS) or 3GPP UMTS; Third Generation Partnership Project 2 (3GPP2) Ultra Mobile Broadband (UMB); High Speed Packet Access (HSPA); High Speed Downlink Packet Access (HSDPA); High Speed Uplink Packet Access (HSUPA); GSM Enhanced Data Rates for GSM Evolution (EDGE) Radio Access Network (RAN) or GERAN; UMTS Terrestrial Radio Access Network (UTRAN); or LTE Advanced.

[0107] What has been described above includes examples of systems and methods illustrative of the disclosed subject matter. It is, of course, not possible to describe every combination of components or methods herein. One of ordinary skill in the art may recognize that many further combinations and permutations of the disclosure are possible. Furthermore, to the extent that the terms “includes,” “has,” “possesses,” and the like are used in the detailed description, claims, appendices and drawings such terms are intended to be inclusive in a manner similar to the term “comprising” as “comprising” is interpreted when employed as a transitional word in a claim.

[0108] While the various embodiments are susceptible to various modifications and alternative constructions, certain illustrated implementations thereof are shown in the drawings and have been described above in detail. It should be understood, however, that there is no intention to limit the various embodiments to the specific forms disclosed, but on the contrary, the intention is to cover all modifications, alternative constructions, and equivalents falling within the spirit and scope of the various embodiments.

[0109] In addition to the various implementations described herein, it is to be understood that other similar implementations can be used, or modifications and additions can be made to the described implementation(s) for performing the same or equivalent function of the corresponding implementation(s) without deviating therefrom. Still further, multiple processing chips or multiple devices can share the performance of one or more functions described herein, and similarly, storage can be affected across a plurality of devices. Accordingly, the embodiments are not to be limited to any single implementation, but rather are to be construed in breadth, spirit and scope in accordance with the appended claims.

What is claimed is:

1. A provisioning system, comprising:

- a processing system including a processor; and
- a memory that stores executable instructions that, when executed by the processing system, facilitate performance of operations, comprising:
 - selecting a logical provisioning template to facilitate a performance of a network provisioning task to provision a network service within a network,
 - based on the logical provisioning template, identifying a programming object comprising logical commands for the performance of the network provisioning task, and
 - executing the logical commands of the programming object to control service implementation equipment to perform the network provisioning task, wherein the logical provisioning template comprises a template usable to provision the network service, wherein the service implementation equipment is part of a network, and wherein the network provisioning task comprises a provisioning of the network

service for equipment that is part of the network, in accordance with the logical provisioning template.

2. The provisioning system of claim 1, wherein the logical commands comprise business logic associated with the network provisioning task independent of commands used by the service implementation equipment.

3. The provisioning system of claim 2, wherein the logical provisioning template comprises a workflow for coordinated performance of processes of the business logic of the network provisioning task.

4. The provisioning system of claim 3, wherein the executing of the programming object controls the service implementation equipment according to the workflow.

5. The provisioning system of claim 1, wherein the programming object is comprised of a non-compiled computer script interpreted by a scripting engine to control the service implementation equipment.

6. The provisioning system of claim 1, wherein the executing of the logical commands of the programming object further controls other service implementation equipment to perform the network provisioning task concurrent with the service implementation equipment performing the network provisioning task.

7. The provisioning system of claim 1, wherein the logical provisioning template further comprises a template usable to provision a label-switching network connection, wherein the service implementation equipment further comprises packet-forwarding equipment that is part of the network, and wherein the network provisioning task comprises a provisioning of the label-switching network connection for other equipment that is part of the network, in accordance with the logical provisioning template.

8. The provisioning system of claim 1, wherein the executing of the logical commands of the programming object comprises executing the logical commands using application data from a data store.

9. The provisioning system of claim 8, wherein the data store comprises the application data for a data-centric performance of the network provisioning task.

10. The provisioning system of claim 8, wherein the executing of the programming object controlling the service implementation equipment, and wherein the operations further comprise, validating the controlling the service implementation equipment based on validation data from the data store.

11. A method, comprising:

facilitating, by a network device comprising a processing system including a processor, selection of a logical provisioning template by an operations device to facilitate a performance of a network provisioning task; and facilitating, by the network device, receiving an indication that service implementation equipment was controlled by the operations device to perform the network provisioning task, wherein the operations device controlled the service implementation equipment based on a programming object comprising logical commands for the performance of the network provisioning task, and wherein the programming object was identified based on the logical provisioning template.

12. The method of claim 11, wherein the logical commands comprise business logic associated with the network provisioning task independent of commands used by the service implementation equipment.

13. The method of claim 12, wherein the logical provisioning template comprises a workflow for coordinated performance of processes of the business logic of the network provisioning task.

14. The method of claim 13, wherein the programming object controls the service implementation equipment according to the workflow.

15. The method of claim 14, wherein the programming object is comprised of a non-compiled computer script interpreted by a scripting engine to control the service implementation equipment.

16. The method of claim 11, wherein an executing of the logical commands of the programming object comprises executing the logical commands using application data from a data store, wherein the logical provisioning template comprises a template usable to provision one of a broadband internet service wherein the service implementation equipment comprises broadband network gateway equipment that is part of a broadband network or a label-switching network connection, and wherein the service implementation equipment comprises packet-forwarding equipment that is part of the broadband network that is part of a broadband network.

17. A non-transitory machine-readable medium, comprising executable instructions that, when executed by a processing system including a processor of network equipment, facilitate performance of operations, the operations comprising:

identifying a programming object comprising logical commands that, when implemented by an operations device, facilitates a performance of a network provisioning task according to a logical provisioning template; and

receiving an indication that service implementation equipment was controlled by the operations device to perform the network provisioning task, wherein the operations device controlled the service implementation equipment based on a programming object comprising logical commands for the performance of the network provisioning task, and wherein the programming object was identified based on the logical provisioning template, wherein the logical provisioning template comprises a template usable to provision one of a network service.

18. The non-transitory machine-readable medium of claim 17, wherein the logical commands comprise business logic associated with the network provisioning task independent of commands used by the service implementation equipment, and wherein the service implementation equipment comprises broadband network gateway equipment that is part of a broadband network or a label-switching network connection, wherein the service implementation equipment comprises packet-forwarding equipment that is part of the broadband network that is part of a broadband network.

19. The non-transitory machine-readable medium of claim 18, wherein the logical provisioning template comprises a workflow for coordinated performance of processes of the business logic of the network provisioning task.

20. The non-transitory machine-readable medium of claim 17, wherein an executing of the logical commands of the programming object controls other service implementation equipment to perform the network provisioning task concurrent with the service implementation equipment performing the network provisioning task.

* * * * *